

# MATH255: Honours Analysis 2

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Abstract

## 1 Metric Spaces

### 1.1 Continuous functions on compact sets

## 2 Infinite Series

**Definition 1** (Partial sum). Let  $(X, \|\cdot\|)$  be a normed vector space over  $\mathbb{R}$  or  $\mathbb{C}$ .  $k_0 \in \mathbb{Z}$  and  $(a_k)_{k \geq k_0}$  by a sequence in  $X$ . For every  $n \geq k_0$ ,  $S_n = \sum_{k=k_0}^n a_k$  is called the partial sum of order  $n$  of  $(a_k)_k$ .

If  $\lim_{n \rightarrow \infty} S_n = S$  in  $(X, \|\cdot\|)$ , for some  $S \in X$ , then we write

$$\sum_{k=k_0}^{\infty} a_k = S \quad (1)$$

and we say that the series  $\sum_{k=k_0}^{\infty} a_k$  and that  $S$  is its sum. If  $(S_n)_{n \geq k_0}$  does not converge, then we say that the series  $\sum_{k=k_0}^{\infty} a_k$  diverges.

### 2.1 Series with nonnegative term

**Theorem 1** (Direct Comparison Test). Let  $\sum_{k=k_0}^{\infty} a_k$  and  $\sum_{k=k_0}^{\infty} b_k$  be two series in  $(\mathbb{R}, \|\cdot\|)$  such that  $0 \leq a_k \leq b_k$  for all  $k \geq k_0$ . Then

1. If  $\sum_{k=k_0}^{\infty} b_k$  converges, then  $\sum_{k=k_0}^{\infty} a_k$  converges and  $\sum_{k=k_0}^{\infty} a_k \leq \sum_{k=k_0}^{\infty} b_k$ .
2. If  $\sum_{k=k_0}^{\infty} a_k$  diverges, then  $\sum_{k=k_0}^{\infty} b_k$  diverges.

**Corollary 1** (Limit Comparison Test). Let  $\sum_{k=k_0}^{\infty} a_k$  and  $\sum_{k=k_0}^{\infty} b_k$  be two series in  $(\mathbb{R}, \|\cdot\|)$  such that  $a_k > 0$  and  $b_k > 0$  for all  $k \geq k_0$ , and  $\lim_{k \rightarrow \infty} \frac{a_k}{b_k} = \ell \in (0, \infty)$ . Then  $\sum_{k=k_0}^{\infty} a_k$  converges if and only if  $\sum_{k=k_0}^{\infty} b_k$  converges.

**Theorem 2** (Root Test).

**Proposition 1.** Let  $\sum_{k=k_0}^{\infty} a_k$  be a series in  $(\mathbb{R}, \|\cdot\|)$  such that  $a_k > 0$  for all  $k \geq k_0$ .

1. If  $\limsup_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$ , then  $\sum_{k=k_0}^{\infty} a_k$  converges.
2. If  $\liminf_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} > 1$ , then  $\sum_{k=k_0}^{\infty} a_k$  diverges.

Proof: (1) Assume that  $R = \limsup_{k \rightarrow \infty} \frac{a_{k+1}}{a_k} < 1$ . Then  $\forall \varepsilon \in (0, 1 - R)$ ,  $\exists n_\varepsilon \in \mathbb{N}$  such that  $\forall k \geq n_\varepsilon$ ,  $\sup_{k \geq n} \frac{a_{k+1}}{a_k} < R + \varepsilon < 1$ , which gives for

all  $k \geq n_\varepsilon$ ,  $a_{k+1} < (R + \varepsilon)a_k < (R + \varepsilon)^{k-n_\varepsilon+1}a_{n_\varepsilon}$ . When  $\varepsilon \in (0, 1 - R)$ ,  $0 < R + \varepsilon < 1$ , so  $\sum_{k=n_\varepsilon}^{\infty} (R + \varepsilon)^{k-n_\varepsilon+1}a_{n_\varepsilon}$  converges, which by direct comparison implies that  $\sum_{k=k_0}^{\infty} a_k$  converges.

(2) Assume that  $R = \liminf \frac{a_{k+1}}{a_k} > 1$ . Then  $\exists k_1 \geq k_0$  such that  $\forall k \geq k_1$ ,  $\inf_{k \geq n} \frac{a_{k+1}}{a_k} > R - \varepsilon > 1$ , i.e.  $a_{k+1} > a_k$ , which implies  $a_k \not\rightarrow 0$  as  $k \rightarrow \infty$ , thus  $\sum_{k=k_0}^{\infty} a_k$  diverges.  $\square$

Now we focus on series in  $(\mathbb{R}, |\cdot|)$  with terms of different sign.

**Theorem 3** (Alternating Series Test). *Let  $\sum_{k=k_0}^{\infty} a_k$  be a series in  $(\mathbb{R}, |\cdot|)$  such that  $0 \leq a_{k+1} \leq a_k$  for all  $k \geq k_0$  and  $\lim_{k \rightarrow \infty} a_k = 0$ . Then  $\sum_{k=k_0}^{\infty} a_k$  converges and*

$$0 \leq \sum_{k=k_0}^{\infty} (-1)^{k-k_0} a_k \leq a_{k_0} \quad (2)$$

### Example 1

$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k}$  converges since  $0 \leq \frac{1}{k+1} \leq \frac{1}{k}$  for all  $k \geq 1$  and  $\lim_{k \rightarrow \infty} \frac{1}{k} = 0$ .

Proof: By observing that

$$\sum_{k=k_0}^{\infty} (-1)^k a_k = \sum_{k=0}^{\infty} (-1)^{k+k_0} a_{k+k_0} \quad (3)$$

$$= (-1)^{k_0} \sum_{k=0}^{\infty} (-1)^k a_{k+k_0} \quad (4)$$

we can assume that  $k_0 = 0$ .

For every  $n \in \mathbb{N}$ ,

$$S_{2n} = a_0 - a_1 + a_2 - \cdots + a_{2n-2} - a_{2n-1} + a_{2n} \quad (5)$$

$$= (a_0 - a_1) + (a_2 - a_3) + \cdots + (a_{2n-2} - a_{2n-1}) + a_{2n} \quad (6)$$

$$\geq 0 \quad (7)$$

$$S_{2n+1} = a_0 - a_1 + a_2 - \cdots + a_{2n-2} - a_{2n-1} + a_{2n} - a_{2n+1} \quad (8)$$

$$= (a_0 - a_1) + (a_2 - a_3) + \cdots + (a_{2n-2} - a_{2n-1}) + (a_{2n} - a_{2n+1}) \quad (9)$$

$$\leq 0 \quad (10)$$

**Definition 2** (Cauchy product). *Let  $\sum_{k=0}^{\infty} a_k$  and  $\sum_{k=0}^{\infty} b_k$  be series in  $(\mathbb{R}, |\cdot|)$  or  $(\mathbb{C}, |\cdot|)$ .*

*For each  $n \in \mathbb{N}$ , define*

$$c_n = \sum_{k=0}^n a_k b_{n-k} = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 \quad (11)$$

*Then the series  $\sum_{n=0}^{\infty} c_n$  is called the Cauchy product of  $\sum_{k=0}^{\infty} a_k$  and  $\sum_{k=0}^{\infty} b_k$ .*

**Proposition 2.** If  $\sum_{k=0}^{\infty} a_k$  and  $\sum_{k=0}^{\infty} b_k$  are series in  $(\mathbb{R}, |\cdot|)$  or  $(\mathbb{C}, |\cdot|)$  such that  $\sum_{k=0}^{\infty} a_k$  converges and  $\sum_{k=0}^{\infty} b_k$  converges (not necessarily absolutely). Then the Cauchy product  $\sum_{n=0}^{\infty} c_n$  converges and  $\sum_{n=0}^{\infty} c_n = (\sum_{k=0}^{\infty} a_k)(\sum_{k=0}^{\infty} b_k)$ .

**Example 2 (Counter example of Cauchy product of two conditionally converging series)**

Consider  $a_k = b_k = \frac{(-1)^k}{\sqrt{k+1}}$ . Then  $\sum_{k=0}^{\infty} a_k$  and  $\sum_{k=0}^{\infty} b_k$  converge by the alternating series test and  $p$ -series convergence.

In this case:

$$|c_n| = \left| \sum_{k=0}^n a_k b_{n-k} \right| \quad (12)$$

$$(13)$$

## References